



Fig. 6: Geometry of textile antenna

are taken at one frequency, one polarisation and one plane cut. Patterns are usually presented in polar or rectilinear form with a dB strength scale. These are normalised to the maximum graph value of 0 dB and directivity is given for the antenna. This means that if side-lobe level from the radiation pattern were down -13 dB, and directivity of the antenna was 4 dB, side-lobe gain would be -9 dB.

Applications of microstrip antennae

Communication-based applications. Microstrip patch antenna finds several applications in wireless communication. For example, satellite communication requires circularly polarised radiation patterns, which can be realised using either square or circular patch microstrip antenna. In global positioning satellite (GPS) systems, circularly polarised microstrip antennae are used. They are very compact in size and quite expensive due to their positioning.

Microstrip antennae are also used in the fields of RFID (radio frequency identification), mobile communication and healthcare. Basically, an RFID system consists of a tag and a reader. Generally, it uses frequencies between 30 Hz and 5.8 GHz.

In telemedicine application, microstrip antennae operate at 2.45 GHz. Wearable microstrip antennae are suitable for wireless body area network. An antenna having gain of 6.7 dB and front-to-back ratio of 11.7 dB, and resonating at 2.45 GHz is suitable for telemedicine applications.

The IEEE 802.16 standard is known

as WiMax (worldwide interoperability for microwave access). It can reach up to 48 km (30-mile) radius with data rate of 70 Mbps. Microstrip antenna can resonate at more than one frequency. Therefore these can be used in WiMax-based communication equipment.

Some communication-based applications of microstrip patch antennae are radio altimeters, command and control systems, remote sensing and environmental instrumentation, feed elements in complex antennae, satellite navigation receivers, mobile radio, integrated antennae, biomedical radiators and intruder alarms, Doppler and other radars, and satellite communication and direct broadcast services.

Mobile communication. Mobile communication requires small, low-cost, low-profile antennae. In some mobile handsets, semiconductor-based diodes or detectors are used as antennae. They are much like p-n diode photodetectors but work at microwave frequency. Many times omnidirectional antenna is used in mobile phones. There are different kinds of antennae like planar inverted-F antenna, folded inverted conformal antenna and monopole. Also retractable whip antenna is commonly used in handsets.

The phone is subdivided into roughly 60 components, each consisting of hundreds or even thousands of individual facets. The internal structure of a mobile handset is shown in Fig. 3.

Medical applications. In the treatment of malignant tumours, microwave energy is said to be the most effective way of inducing hyperthermia. The radiator to be used for this

purpose should be light-weight, easy to handle and rugged. Only a patch radiator fulfils these requirements.

The initial designs of microstrip radiators for inducing hyperthermia were based on printed dipoles and annular rings that were designed on S-band (2-4 GHz). Later on the design was based on a circular microstrip disk at L-band (1-2 GHz). Two coupled microstrip lines with a flexible separation are used to measure temperature inside the human body. Fig. 5 shows a flexible patch applicator that operates at 430 MHz.

Textile antennae: recent research

There are some applications at present where antennae are used to continuously monitor biometric data of the human body. In order to do this, they need to be so close to the human body all the time that they can continuously monitor the biometric data and send the information to the outside world. If the antenna is hard, it cannot be kept always attached with the human body. An antenna made of textile material will not harm the human body and can be worn for extended periods. Wearable antennae will find use in healthcare, recreation, fire-fighting, etc.

Textile materials are increasingly being used for development of flexible wearable systems due to the recent miniaturisation of wireless devices. For flexible antennae, textile materials form interesting substrates, because fabric antennae can be easily integrated into clothes.

In this special type of patch antenna, the radiating patch and the ground plane are made up of conductive textile material. The substrate too is a textile material with specific dielectric constant. As everything is made up of textile material, it is called textile antenna.

Textile antennae are still under testing, so they are rarely in focus. ●

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